Fundamental Concepts Exercises

1. What is a database?

A. A program used only to create spreadsheets
B. An organized, shared collection of related data that supports an organization's activities
C. A language used to write queries
D. A collection of unrelated files

2. What is a central advantage of a shared database over separate, isolated files?

A. Different users and applications can access related data in one organized collection.
B. Every application must maintain its own copy of all data.
C. Users no longer need access controls.
D. All information must be stored in one table.

3. What is a relational database?

A. A single file containing unstructured text
B. A collection of application programs
C. Software used to manage user accounts
D. A collection of related tables

4. What is a possible connection between Employee and Department tables?

A. Employee names
B. The order of their rows
C. DepartmentID
D. Their filenames

5. What is the difference between a database and a DBMS?

A. A database processes queries, while a DBMS stores only table names.
B. A database contains related data, while a DBMS is the software that manages it.
C. A database is a programming language, while a DBMS is a table.
D. Database and DBMS are two names for exactly the same thing.

6. Which of the following is an example of a DBMS?

A. MySQL
B. SQL
C. An Employee table
D. A customer record

7. Why is a DBMS described as a "collection of programs"?

A. It stores every application installed on a computer.
B. It requires a separate database for every SQL command.
C. It performs only one operation at a time.
D. It includes components for tasks such as querying, security, transactions, and recovery.

8. Which statement correctly describes SQL?

A. SQL is the database containing an organization's records.
B. SQL is another name for MySQL.
C. SQL is a language used to communicate with a relational DBMS.
D. SQL is an application used only by bank tellers.

9. What happens when you execute this statement?

```
SELECT *
FROM Employee;
```

A. SQL independently accesses the data without a DBMS.
B. The DBMS interprets the statement, retrieves the requested data, and returns the results.
C. The Employee table is deleted.
D. A separate database is created for each employee.

10. How does a banking application typically access stored data?

A. The application sends requests to the DBMS, which accesses the database.
B. The application requires the teller to write every SQL statement manually.
C. The database sends requests to the teller, who forwards them to the DBMS.
D. Each banking application must use a different database.